

```
!pip install openai google-generativeai transformers torch --quiet
```

```
from openai import OpenAI
import google.generativeai as genai
from transformers import AutoModelForCausalLM, AutoTokenizer
from IPython.display import Markdown, display
import torch
import json
import os
```

```
os.environ["OPENAI_API_KEY"] = input("Enter OpenAI API Key: ")
os.environ["GEMINI_API_KEY"] = input("Enter Gemini API Key: ")
```

```
client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])
genai.configure(api_key=os.environ["GEMINI_API_KEY"])
```

```
Enter OpenAI API Key: sk-proj-MHb7WT0IMkkyoW61RP7VUTbTtCdFat-q0ksSHPggRptQW9Y_9BHDY-WsBuMwsrzZoqW5iSXN2RT3B1bkFJX4-5C9KMn77_
Enter Gemini API Key: AIzaSyC-wti3_DwmqIzHhTJs5pLbyDQ25wNYlro
```

```
model_name = "distilgpt2" # smaller & faster
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)
```

```
print("Hugging Face Model Loaded")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
```

```
config.json: 100% 762/762 [00:00<00:00, 67.0kB/s]
```

```
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 2.41kB/s]
```

```
vocab.json: 100% 1.04M/1.04M [00:00<00:00, 1.90MB/s]
```

```
merges.txt: 100% 456k/456k [00:00<00:00, 1.62MB/s]
```

```
tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 1.69MB/s]
```

```
model.safetensors: 100% 353M/353M [00:03<00:00, 193MB/s]
```

```
Loading weights: 100% 76/76 [00:00<00:00, 391.89it/s, Materializing param=transformer.wte.weight]
```

```
GPT2LMHeadModel LOAD REPORT from: distilgpt2
```

```
Key | Status | |
-----+-----+--
transformer.h.{0, 1, 2, 3, 4, 5}.attn.bias | UNEXPECTED | |
```

```
Notes:
```

```
- UNEXPECTED : can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
generation_config.json: 100% 124/124 [00:00<00:00, 15.0kB/s]
```

```
Hugging Face Model Loaded
```

```
# OpenAI (Safe Version)
```

```
def generate_openai(prompt):
```

```
    try:
```

```
        response = client.chat.completions.create(
            model="gpt-4o-mini",
            messages=[{"role": "user", "content": prompt}],
            max_tokens=100
        )
        return response.choices[0].message.content.strip()
```

```
    except Exception as e:
```

```
        return f"OpenAI Error: {str(e)}"
```

```
# Gemini
```

```
def generate_gemini(prompt):
```

```
    try:
```

```
        gemini_model = genai.GenerativeModel("gemini-1.5-flash")
        response = gemini_model.generate_content(prompt)
        return response.text.strip()
```

```
    except Exception as e:
```

```
        return f"Gemini Error: {str(e)}"
```

```
# Hugging Face
```

```
def generate_hf(prompt):
    try:
        inputs = tokenizer(prompt, return_tensors="pt")
        outputs = model.generate(
            inputs.input_ids,
            max_length=100,
            do_sample=True,
            temperature=0.7
        )
        return tokenizer.decode(outputs[0], skip_special_tokens=True)

    except Exception as e:
        return f"HuggingFace Error: {str(e)}"
```

```
prompt = "Write a haiku about AI and nature."
```

```
competitors = ["OpenAI", "Gemini", "Hugging Face"]
answers = []
```

```
answers.append(generate_openai(prompt))
answers.append(generate_gemini(prompt))
answers.append(generate_hf(prompt))
```

WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint (:::1) 659. The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass you Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation. The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you

```
for comp, ans in zip(competitors, answers):
    display(Markdown(f"## {comp} Output:\n{ans}\n---"))
```

OpenAI Output:

OpenAI Error: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, read the docs: <https://platform.openai.com/docs/guides/error-codes/api-errors>.'}, 'type': 'insufficient_quota', 'param': None, 'code': 'insufficient_quota'}}

Gemini Output:

Gemini Error: 404 POST <https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint>: models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ListModels to see the list of available models and their supported methods.

Hugging Face Output:

Write a haiku about AI and nature.

The idea of a virtual world is so important to us that we think of the world as a whole, not a unit. But if, for example, you are a robot or a machine learning machine, you could create a virtual world of objects. The idea of a virtual world is so important to us that we think of the world as a whole, not a unit. But if, for example, you are a robot or

```
judge_prompt = f"""
Rank the following model outputs based on creativity, clarity, and relevance.
```

```
1. {answers[0]}
2. {answers[1]}
3. {answers[2]}
```

```
Return JSON only:
{"results": ["1", "2", "3"]}
"""
```

```
judge_response = generate_gemini(judge_prompt)
```

```
print("Judge Response:")
print(judge_response)
```

```
try:
    ranking = json.loads(judge_response)["results"]
    ranked_models = [competitors[int(r)-1] for r in ranking]
```

```
print("\nFinal Ranking (Best → Worst):")
for i, model in enumerate(ranked_models, 1):
    print(f"Rank {i}: {model}")

except:
    print("Ranking could not be parsed automatically.")
```

WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint (::1) 279.
Judge Response:
Gemini Error: 404 POST <https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?%24alt=json?>
Ranking could not be parsed automatically.